Engine Noise Diagnostic Procedures - General Information

Note : When diagnosing engine noise problems, make sure that noises caused by accessories, such as the air compressor and power takeoff, are not mistaken for engine noises. Remove the accessory drive belts to eliminate noise caused by these units. Noise will also travel to other metal parts not related to the problem. The use of a stethoscope can help locate an engine noise.

Engine noises heard at the crankshaft speed, engine rpm, are noises related to the crankshaft, rods, pistons, and piston pins. Noises heard at the camshaft speed, one-half of the engine rpm, are related to the valve train. A hand-held digital tachometer can help to determine if the noise is related to components operating at the crankshaft or camshaft speed.

Engine noise can sometimes be isolated by performing a cylinder cutout test. Refer to Procedure 014-005 (Engine Testing (Engine Dynamometer) in Section 14. (/qs3/pubsys2/xml/en/procedures/20/20-014-005.html) If the volume of the noise decreases or the noise disappears, it is related to that particular engine cylinder.

There is **not** a definite rule or test that will positively determine the source of a noise complaint.

Engine driven components and accessories, such as gear-driven fan clutches, hydraulic pumps, belt-driven alternators, air-conditioning compressors, and turbochargers can contribute to engine noise. Use the following information as a guide to diagnosing engine noise.

Main Bearing Noise

Refer to Engine Noise Excessive - Main Bearing symptom tree.

The noise caused by a loose main bearing is a loud dull knock heard when the engine is pulling a load. If all main bearings are loose, a loud clatter will be heard. The knock is heard regularly every other revolution. The noise is the loudest when the engine is lugging or under heavy load. The knock is duller than a connecting rod noise. Low oil pressure can also accompany this condition.

If the bearing is **not** loose enough to produce a knock by itself, the bearing can knock if the oil is too thin, or if there is no oil at the bearing.

An irregular noise can indicate worn crankshaft thrust bearings.

An intermittent sharp knock indicates excessive crankshaft end clearance. Repeated clutch disengagements can cause a change in the noise.

Connecting Rod Bearing Noise

Refer to Engine Noise Excessive - Piston symptom tree.

It is difficult to tell the difference between piston pin, connecting rod, and piston noise. A loose piston pin causes a loud double knock which is usually heard when the engine is idling. When the injector to this cylinder is cut out, a noticeable change will be heard in the sound of the knocking noise. However, on some engines the knock becomes more noticeable when the vehicle is operated on the road at steady speed condition.

Piston Noise

Refer to Engine Noise Excessive - Connecting Rod symptom tree.

Connecting rods with excessive clearance knock at all engine speeds, and under both idle and load conditions. When the bearings begin to become loose, the noise can be confused with piston slap or loose piston pins. The noise increases in volume with engine speed. Low oil pressure can also accompany this condition.

Driveability - General Information

Driveability is a term that generally describes vehicle performance on the road. Driveability problems for an engine can be caused by several different factors. Some of the factors are engine related and some are not.

Before troubleshooting, it is important to determine the exact complaint and whether the engine has a real driveability problem, or if it simply does **not** meet driver expectations. The “Driveability-Low Power Customer Complaint Form” is a valuable list of questions that **must** be used to assist the service technician in determining what type of driveability problem the vehicle is experiencing. The checklist **must** be completed before troubleshooting the problem. The form can be found at the end of this section. If an engine is performing to factory specifications but does **not** meet customer expectations, it **must** be explained to the customer that nothing is wrong with the engine and why.

The troubleshooting symptom charts have been set up to divide driveability problems into two different symptoms: “Engine Power Output Low” and “Engine Acceleration or Response Poor”.

Low power is a term that is used in the field to describe many different performance problems. However, in this manual low power is defined as the inability of the engine to produce the power necessary to move the vehicle at a speed that can be reasonably expected under the given conditions of load, grade, wind, and so on. Low power is usually caused by the lack of fuel flow which can be caused by any of the following factors:

- Lack of full travel of the throttle pedal
- Failed boost sensor
- Excessive fuel inlet, intake, exhaust, or drainline restriction
- Loose fuel pump suction lines

Low power is **not** the inability of the vehicle to accelerate satisfactorily from a stop or the bottom of a grade. Refer to the symptom tree "Engine Power Output Low" for the proper procedures to locate and correct a low power problem. The chart starts off with basic items which can cause lower power. It then breaks off into application specific items which can cause low power. All of the "Application Specific" charts end with a step called "Fuel or Air Delivery Problem". This step leads to an engine performance check that requires engine measurements. The last section of this chart is titled "Performance Measurement" which leads the troubleshooter through the causes and corrections based on the outcome of the performance check.

Poor acceleration or response is described in this manual as the inability of the vehicle to accelerate satisfactorily from a stop or from the bottom of a grade. It can also be the lag in acceleration during an attempt to pass or overtake another vehicle at conditions less than rated speed and load. Poor acceleration or response is difficult to troubleshoot since it can be caused by factors such as:

- Engine or pump related factors
- Driver technique
- Improper gearing
- Improper engine application
- Worn clutch or clutch linkage

Engine related poor acceleration or response can be caused by several different factors such as:

- Failed boost sensor
- Excessive drainline restriction
- Throttle deadband

Refer to the symptom tree Engine Acceleration or Response Poor for the proper procedures to locate and correct a poor acceleration or response complaint.

Driveability/Low Power - Customer Complaint Form

Customer Name/Company ______________________________ Date ____________

How did the problem occur?

Suddenly _______ Gradually _______At what hour/mileage did the problem begin?
Hours _______ Miles _______ Since New _______

- **After engine repair?**
- **Yes _______ No _______ After equipment repair?**
- **Yes _______ No _______ After change in equipment use?**
- **Yes _______ No _______ After change in selectable programmable parameters?**
- **Yes _______ No _______ If so, what was repaired and when?**

  ____
- **Does the vehicle also experience poor fuel economy? Yes _______ No _______.**

Answer questions 4 through 8 using selections (A through F) listed below. Circle the letter or letters that best describes the complaint.

A - Compared to fleet
B - Compared to competition
C - Compared to previous engine
D - Personal expectation
E - Won't pull on hill
F - Won't pull on flat

A B C D E F

Can the vehicle obtain the expected road speed? Yes _______ No _______
What is desired speed? rpm/mph _______
What is achieved speed? rpm/mph _______
GVW ______________

A B C D

Is the vehicle able to pull the load? Yes _______ No _______

When?

_______ In the hills
_______ With a loaded trailer
_______ On the flat
Other____

If no was the answer to the previous questions, fill out the Driveability/Low Power/Excessive Fuel Consumption Checklist and go to the Low Power performance tree

A B C D E F

Is the vehicle slow to accelerate or respond? Yes _______ No _______
From a stop? Yes _______ No _______
After a shift? Yes _______ No _______ rpm _______
Before a shift? Yes _______ No _______ rpm _______
No shift? Yes _______ No _______ rpm _______

A B C D

Does the vehicle hesitate after periods of long deceleration or coasting? Yes _______ No _______ rpm

If yes was the answer to the previous two questions, fill out the Driveability/Low Power/ Excessive Fuel Consumption Checklist and go to the Poor Acceleration/Response performance tree.

A B C D E F

Additional Comments:

____

____

____

This page can be copied for your convenience.

Fuel Consumption - General Information

The cause of excessive fuel consumption is hard to diagnose and correct because of the potential number of factors involved. Actual fuel consumption problems can be caused by any of the following factors:

Engine factors
Vehicle factors and specifications
Environmental factors
Driver technique and operating practices
Fuel system factors
Low power/driveability problems

Before troubleshooting, it is important to determine the exact complaint. Is the complaint based on whether the problem is real or perceived, or does **not** meet driver expectations? The “Fuel Consumption - Customer Complaint Form” (on the next page) is a valuable list of questions that can be used to assist the service technician in determining the cause of the problem. Complete the form before troubleshooting the complaint. The following are some of the factors that **must** be considered when troubleshooting fuel consumption complaints.

- An operator will change driving style to compensate for a low power/driveability problem. Some things the driver is likely to do are, (a) shift to a higher engine rpm or (b) run on the droop curve in a lower gear instead of upshifting to drive at part throttle conditions. These changes in driving style will increase the amount of fuel used.
- As a general rule, a 1.6 km [1 mph] increase in road speed equals a 0.16 km [0.1 mpg] increase in fuel consumption. This means that increasing road speed from 50 to 60 mph will

result in a loss of fuel mileage of 1.6 km [1 mpg].

- As a general rule, there can be as much as a 1.6 to 2.4 km [1 to 1.5 mpg] difference in fuel consumption depending on the season and the weather conditions.
- Idling the engine can use from 0.11 to 0.33 liters [0.5 to 1.5 gal] per hour depending on the engine idle speed.
- East/west routes experience almost continual crosswinds and head winds. Less fuel can be used on north/south routes where parts of the trip are **not** only warmer, but see less wind resistance.

Additional vehicle factors, vehicle specifications, and axle alignment can also affect fuel consumption. For additional information on troubleshooting fuel consumption complaints, refer to Fuels for Cummins Engines, Bulletin 3379001. (/qs3/pubsys2/xml/en/bulletin/3379001.html)

Fuel Consumption - Customer Complaint Form

Customer Name/Company ____ Date ____________

Answer the following questions. Some questions require making an X next to the appropriate answer.

1. What fuel mileage is expected? ______ Expected mpg
2. What are the expectations based on? Original mileage ______, Other units in fleet ______, Competitive engines ______Previous engine owned ______, Expectations only ______, VE/VMS report ______
3. When did the problem occur? Since New ______, Suddenly ______, Gradually ______
4. Did the problem start after a repair? Yes ______ No ______ If so, what was repaired and when? ____
5. Is the vehicle also experiencing a Driveability problem (Low Power or Poor Acceleration/Response)? Yes ______ No ______

If answered Yes, fill out the Driveability/Low Power/Excessive Fuel Consumption Checklist and go to the Engine Power Output Low performance tree.

6. Is the problem seasonal? Yes ______ No ______
7. Weather conditions during fuel consumption check? Rain ____, Snow ____, Windy ____, Hot Temperatures ____, Cold Temperatures ____
8. How is the fuel mileage measured? Tank ____, Trip ____, Month ____, Year ____ Hubometer ____, Odometer ____
9. Are accurate records kept of fuel added on the road? Yes ______ No ______
10. Do routes vary between compared vehicles? Yes ______ No ______
11. Have routes changed for the engine being checked? Yes ______ No ______
12. What are the loads hauled, compared to comparison unit? GVW ____________ Heavier ______, Lighter ______

13. What is the altitude during operation? Below 10,000 feet ________, Above 10,000 feet ________
14. How much of the time is the truck spent idling? Hours/day __________
15. Is the driver technique or operating practices affecting fuel economy?

- High road speed: mph __________
- Operate at rated speed or above: rpm __________
- Incorrect shift rpm: Shift rpm __________, Torque Peak __________
- Operate at a cruise speed: rpm __________
- Believe compensating for low power: Yes ________ No _______

If after filling out this form it appears that the problem is **not** caused by vehicle factors, environmental factors, or driver technique, fill out the Driveability/Low Power/Excessive Fuel Consumption Checklist and go to the Fuel Consumption Excessive performance tree.

This form can be copied for convenience.

Oil Consumption

In addition to the information that follows, a service publication is available entitled Technical Overview of Oil Consumption, Bulletin 3379214.

Cummins Inc. defines "Acceptable Oil Usage" as outlined in the following table.

Acceptable Oil Usage									
Any Time During Coverage Period									
Engine Family	Hours per Quart	Hours per Liter	Hours per Imperial Quart	Miles Per Quart	Miles per Liter	Miles per Imperial Quart	KM per Quart	KM per Liter	KM per Imperial Liter
A	10.0	10.6	12.0	400	425	475	650	675	775
4B	10.0	10.6	12.0	400	425	475	650	675	775
6B	10.0	10.6	12.0	400	425	475	650	675	775
6C	10.0	10.6	12.0	400	425	475	650	675	775
V/VT-378	4.0	4.3	5.0	-	-	-	-	-	-
V/VT-504	4.0	4.3	5.0	250	265	310	400	425	485
V/VT-555	4.0	4.3	5.0	250	265	310	400	425	485
L Series	4.0	4.3	5.0	250	265	310	400	425	485

M Series	4.0	4.3	5.0	250	265	310	400	425	485
N Series	4.0	4.3	5.0	250	265	310	400	425	485
V/VT/ VTA-903	4.0	4.3	5.0	250	265	310	400	425	485
KT/KTA-19	3.0	3.2	3.75	200	210	250	320	340	390
QSK23	1.7	1.8	2.0	-	-	-	-	-	-
V/VT/ VTA28	2.0	2.1	2.5	-	-	-	-	-	-
KT/KTA38	1.5	1.6	1.8	-	-	-	-	-	-
KTA50	1.1	1.2	1.3	-	-	-	-	-	-

Acceptable Oil Useage (Transit Bus, Shuttle Bus and School Bus)									
Any Time During Coverage Period									
Engine Family	Hours per Quart	Hours per Liter	Hours per Imperial Quart	Miles per Quart	Miles per Liter	Miles per Imperial Quart	KM per Quart	KM per Liter	KM per Imperial Liter
B	10.0	10.6	12.0	200	210	240	320	340	385
C	8.0	8.5	10.0	150	160	180	240	255	290
L, M, N	4.0	4.3	5.0	100	105	120	160	170	195

Cummins Engine Company, Inc.
Box 3005
Columbus, IN, U.S.A.
47202-3005

15200020

Engine Lubricating Oil Consumption Report				
Owner's Name	Date of Delivery			Engine Serial Number
	Month	Day	Year	
Address	Equipment Manufacturer			Engine Model & HP

City	State/Province	Equipment Serial Number	Fuel Pump Serial Number	
Engine Application (Describe)	Oil and Filter Change Interval		Complaint Originally Registered	
	Oil	Filters	Date	Mile/Hours/KM
Lubricating Oil Added				
Date Added Oil	Engine Operation Miles/Hours/Kilometers	Quarts - Liters Oil Added	Brand and Viscosity of Oil Used	
Start Test				
Last Mileage/Hours/Kilometers ____________ Minus Start Mileage/Hours/Kilometers ____________				
Equals Test Mileage/Hours/Kilometers ____________ Divided By Oil Added ____________				
Equals ____________ Usage Rate____________				
Customer Signature	Cummins Dealer		Cummins Distributor	
Cummins Inc. Form 4755				

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Columbus, IN, U.S.A.
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Oil Consumption Report	
Customer Name:	Dist/Dir:
Engine Model:	Mi/Km/Hr:
Engine Serial No.:	CPL No.:
Vehicle Make/Model:	Date:
Review of maintenance history: List any previous failures that could have had a detrimental effect on cylinder component life. Failures could include fuel, coolant, and/or foreign abrasives in the oil, second ring groove beat-out, filter plugging, etc.	
Lube Oil Used: Brand Viscosity Change Interval (mi/km/hr)	
Combination Oil Filter: Model Element Change Interval (mi/km/hr)	
Bypass Oil Filter: Model Element Change Interval (mi/km/hr)	

Full Flow Oil Filter: Model Element Change Interval (mi/km/hr)
Air Cleaner: Make and Model Change Interval
List any external engine leaks.
Visually check for any internal leaks and list. Check turbocharger seals, valve guides, air compressor, etc.
Had the fuel pump been tampered with? __________ What is maximum rail pressure readings? __________ If yes, the pump must be reset to factory specifications and the customer sent out to re-evalute his oil consumption rate and the eligibility requirements must be met again.
Warning: Governmental agencies have determined that used engine oil is toxic and carcinogenic. Avoid breathing, injestion, and excessive contact. Drain and refill oil pan to check dipstick markings and notes findings.
Only after above checks are completed, leaks corrected and proper documentation is completed, disassemble engine to determine cause for the failure and repair as required.
State reason for oil consumption.
Signed: ____

Coolant Loss Pre-Troubleshooting Guide

Before troubleshooting, it is critical to know where the coolant is being lost. It is **not** always obvious where the missing coolant has gone.

Before troubleshooting, it is important to determine the exact complaint by interviewing the driver, looking at the service history and looking at the ECM information.

Driver Interview Questions

Driver's Name:

Engine Serial Number (ESN):

What is the complaint?

How is this engine used?

What are the load factors?

Where is the vehicle driven?

1. How is radiator filled?
2. Is the radiator filled to the High or Low mark when the engine is cold (less than 60°C [140°F]?
3. What type of coolant is added?
4. Is there been any coolant on the ground under your vehicle or engine?
5. Is there green or white streaks on the engine or near the coolant overflow hose?

6. Is there any specific condition when you get indications of coolant loss (weather, altitude, or load)?
7. Does the engine ever overheat?
8. Does the warning light flash?
9. Under what conditions?
10. What temperature does the coolant run at normally?
11. Does the cooling fan operate correctly (fan on a 99°C [210°F]?
12. Is there been any white smoke at operating temperature, or has anyone told you that white smoke is coming out of the exhaust?
13. Is oil analysis performed as part of the maintenance?
14. Are there elevated levels of sodium or potassium?
15. Has any increase in moisture condensation on the dipstick or oil fill cap, or moisture in the blowby been evident?
16. Has a milky appearance been noticed in the lube oil that might indicate coolant is present.
17. What other comments did the driver/operator have that might help Cummins Inc. make the right repair?

Service History Review

Repeat cylinder head or cylinder head gasket repairs can indicate the problem is likely **not** the cylinder head or cylinder head gasket. Repeat problems can indicate a deeper problem in the engine. Keep this information in mind while going through the troubleshooting procedure.

Look at this engine's warranty claims history: who worked on engine last and what did technician do? How many miles miles/kilometers are on the engine? Has a cylinder head or cylinder head gasket been replaced before? At what engine hours were these repairs made?

ECM Data Review

Print out an INSITE™ electronic service tool Image Report from the ECM. Look for high temperature alarms or low coolant level alarms. Either indication confirms a complaint of loosing coolant.

Are any fault codes logged in the Engine Protection Fault History?

ECM Fault Code 235 - Low Coolant (how may times)?

ECM Fault Code 151 - High Coolant Temperature (how many times)?

At this point, do you know where the coolant is going? If not and the coolant loss is not severe, suggest mounting a catch bottle on the radiator overflow tube to catch any overflow that can possibly be blowing out and becoming lost while at speed. Send the vehicle out to collect more data about where the coolant is or is not going. If the catch bottle has some coolant in it, refer back to the Coolant Loss External (out the overflow) interview questions.

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